MongoDB Lab Script: Telecom Use Case

Scenario

Telecom company wants to manage:

Subscribers

Plans

Recharge Transactions

Call Detail Records

Network Events

Billing

1. Create Database

```
> use("telecom_lab_db");
< switched to db telecom_lab_db
```

2. Drop Old Collections

```
> db.subscribers.drop();
  db.plans.drop();
  db.recharges.drop();
  db.call_records.drop();
  db.network_events.drop();
  db.billing.drop();
< true
```

3. Insert Subscribers

```
>_MONGOSH
> db.subscribers.insertMany([
    {
      subscriber_id: 1001,
      msisdn: "959420001001",
      name: "Aung Aung",
      city: "Yangon",
      township: "Hlaing",
      sim_type: "Prepaid",
      status: "Active",
      plan_id: "P001",
      activation_date: ISODate("2025-01-10")
    },
    {
      subscriber_id: 1002,
      msisdn: "959420001002",
      name: "Su Su",
      city: "Mandalay",
      township: "Chan Aye Thar Zan",
      sim_type: "Postpaid",
      status: "Active",
      plan_id: "P002",
      activation_date: ISODate("2024-11-15")
    },
    {
      subscriber_id: 1003,
      msisdn: "959420001003",
      name: "Mg Mg",
      city: "Naypyidaw",
      township: "Zabuthiri",
      sim_type: "Prepaid",
      status: "Suspended",
      plan_id: "P003",
      activation_date: ISODate("2025-03-01")
    },
    {
      subscriber_id: 1004,
      msisdn: "959420001004",
      name: "Hla Hla",
      city: "Yangon",
      township: "Tamwe",
      sim_type: "Prepaid",
      status: "Active",
      plan_id: "P001",
      activation_date: ISODate("2025-02-20")
    }
  ]);
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6ace1dd34886783d50a832cd'),
      '1': ObjectId('6ace1dd34886783d50a832ce'),
      '2': ObjectId('6ace1dd34886783d50a832cf'),
      '3': ObjectId('6ace1dd34886783d50a832d0')
    }
  }
```

4. Insert Plans

```
>_MONGOSH
> db.plans.insertMany([
    {
      plan_id: "P001",
      plan_name: "Data Starter",
      monthly_fee: 5000,
      data_gb: 5,
      voice_minutes: 100,
      sms: 50
    },
    {
      plan_id: "P002",
      plan_name: "Business Plus",
      monthly_fee: 30000,
      data_gb: 50,
      voice_minutes: 1000,
      sms: 500
    },
    {
      plan_id: "P003",
      plan_name: "Youth Pack",
      monthly_fee: 10000,
      data_gb: 15,
      voice_minutes: 300,
      sms: 100
    }
  ]);
```

```
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6ace1e484886783d50a832d1'),
      '1': ObjectId('6ace1e484886783d50a832d2'),
      '2': ObjectId('6ace1e484886783d50a832d3')
    }
  }
```

5. Insert Recharge Transactions

```
>_MONGOSH
> db.recharges.insertMany([
    {
      recharge_id: "R001",
      subscriber_id: 1001,
      amount: 5000,
      channel: "Mobile App",
      recharge_date: ISODate("2026-07-01T09:10:00Z"),
      status: "Success"
    },
    {
      recharge_id: "R002",
      subscriber_id: 1001,
      amount: 3000,
      channel: "Retail Shop",
      recharge_date: ISODate("2026-07-02T12:30:00Z"),
      status: "Success"
    },
    {
      recharge_id: "R003",
      subscriber_id: 1003,
      amount: 1000,
      channel: "USSD",
      recharge_date: ISODate("2026-07-02T18:45:00Z"),
      status: "Failed"
    },
    {
      recharge_id: "R004",
      subscriber_id: 1004,
      amount: 10000,
      channel: "Mobile App",
      recharge_date: ISODate("2026-07-03T08:20:00Z"),
      status: "Success"
    }
  ]);
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6ace1f424886783d50a832d4'),
      '1': ObjectId('6ace1f424886783d50a832d5'),
      '2': ObjectId('6ace1f424886783d50a832d6'),
      '3': ObjectId('6ace1f424886783d50a832d7')
    }
  }
```

6. Insert Call Detail Records

>_MONGOSH

```
> db.call_records.insertMany([
    {
      cdr_id: "CDR001",
      caller_msisdn: "959420001001",
      receiver_msisdn: "959420001002",
      call_type: "Voice",
      duration_sec: 180,
      cell_tower_id: "T001",
      call_start: ISODate("2026-07-01T10:00:00Z"),
      cost: 120
    },
    {
      cdr_id: "CDR002",
      caller_msisdn: "959420001002",
      receiver_msisdn: "959420001001",
      call_type: "Voice",
      duration_sec: 600,
      cell_tower_id: "T002",
      call_start: ISODate("2026-07-01T11:15:00Z"),
      cost: 350
    },
    {
      cdr_id: "CDR003",
      caller_msisdn: "959420001004",
      receiver_msisdn: "959420001003",
      call_type: "Voice",
      duration_sec: 90,
      cell_tower_id: "T001",
      call_start: ISODate("2026-07-02T15:20:00Z"),
      cost: 80
    },
    {
      cdr_id: "CDR004",
      caller_msisdn: "959420001001",
      receiver_msisdn: "959420001004",
      call_type: "Data",
      data_mb: 850,
      cell_tower_id: "T003",
      call_start: ISODate("2026-07-03T20:10:00Z"),
      cost: 450
    }
]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ace1fdd4886783d50a832d8'),
    '1': ObjectId('6ace1fdd4886783d50a832d9'),
    '2': ObjectId('6ace1fdd4886783d50a832da'),
    '3': ObjectId('6ace1fdd4886783d50a832db')
  }
}
```

7. Insert Network Events

```
>_MONGOSH
> db.network_events.insertMany([
    {
      event_id: "N001",
      cell_tower_id: "T001",
      city: "Yangon",
      event_type: "Congestion",
      severity: "High",
      affected_users: 1200,
      event_time: ISODate("2026-07-01T18:00:00Z"),
      resolved: false
    },
    {
      event_id: "N002",
      cell_tower_id: "T002",
      city: "Mandalay",
      event_type: "Power Failure",
      severity: "Critical",
      affected_users: 3000,
      event_time: ISODate("2026-07-02T02:30:00Z"),
      resolved: true
    },
    {
      event_id: "N003",
      cell_tower_id: "T003",
      city: "Yangon",
      event_type: "Packet Loss",
      severity: "Medium",
      affected_users: 700,
      event_time: ISODate("2026-07-03T21:00:00Z"),
      resolved: false
    }
  ]);
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6ace20b34886783d50a832dc'),
      '1': ObjectId('6ace20b34886783d50a832dd'),
      '2': ObjectId('6ace20b34886783d50a832de')
    }
  }
```

8. Insert Billing

```
>_MONGOSH
> db.billing.insertMany([
    {
      bill_id: "B001",
      subscriber_id: 1002,
      billing_month: "2026-07",
      monthly_fee: 30000,
      usage_charge: 12000,
      tax: 2100,
      total_amount: 44100,
      payment_status: "Unpaid"
    },
    {
      bill_id: "B002",
      subscriber_id: 1001,
      billing_month: "2026-07",
      monthly_fee: 5000,
      usage_charge: 3500,
      tax: 425,
      total_amount: 8925,
      payment_status: "Paid"
    }
  ]);
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('6ace21904886783d50a832df'),
      '1': ObjectId('6ace21904886783d50a832e0')
    }
  }
```

Lab Exercises

Exercise 1: Find Active Prepaid Subscribers

```
> db.subscribers.find({
    sim_type: "Prepaid",
    status: "Active"
  });
< {
    _id: ObjectId('6ace1dd34886783d50a832cd'),
    subscriber_id: 1001,
    msisdn: '959420001001',
    name: 'Aung Aung',
    city: 'Yangon',
    township: 'Hlaing',
    sim_type: 'Prepaid',
    status: 'Active',
    plan_id: 'P001',
    activation_date: ISODate('2025-01-10T00:00:00.000Z')
  }
  {
    _id: ObjectId('6ace1dd34886783d50a832d0'),
    subscriber_id: 1004,
    msisdn: '959420001004',
    name: 'Hla Hla',
    city: 'Yangon',
    township: 'Tamwe',
    sim_type: 'Prepaid',
    status: 'Active',
    plan_id: 'P001',
    activation_date: ISODate('2025-02-20T00:00:00.000Z')
  }
```

Exercise 2: Find High Recharge Customers

```
> db.recharges.find({
    amount: { $gte: 5000 },
    status: "Success"
  });
< {
    _id: ObjectId('6ace1f424886783d50a832d4'),
    recharge_id: 'R001',
    subscriber_id: 1001,
    amount: 5000,
    channel: 'Mobile App',
    recharge_date: ISODate('2026-07-01T09:10:00.000Z'),
    status: 'Success'
  }
  {
    _id: ObjectId('6ace1f424886783d50a832d7'),
    recharge_id: 'R004',
    subscriber_id: 1004,
    amount: 10000,
    channel: 'Mobile App',
    recharge_date: ISODate('2026-07-03T08:20:00.000Z'),
    status: 'Success'
  }
```

Exercise 3: Total Recharge Amount by Subscriber

```
> db.recharges.aggregate([
    { $match: { status: "Success" } },
    {
      $group: {
        _id: "$subscriber_id",
        total_recharge: { $sum: "$amount" },
        recharge_count: { $sum: 1 }
      }
    },
    { $sort: { total_recharge: -1 } }
  ]);
< {
    _id: 1004,
    total_recharge: 10000,
    recharge_count: 1
  }
  {
    _id: 1001,
    total_recharge: 8000,
    recharge_count: 2
  }
```

Exercise 4: Join Subscribers with Plans

>_MONGOSH

```
db.subscribers.aggregate([
  {
    $lookup: {
      from: "plans",
      localField: "plan_id",
      foreignField: "plan_id",
      as: "plan_info"
    }
  },
  { $unwind: "$plan_info" },
  {
    $project: {
      _id: 0,
      subscriber_id: 1,
      name: 1,
      msisdn: 1,
      city: 1,
      plan_name: "$plan_info.plan_name",
      monthly_fee: "$plan_info.monthly_fee",
      data_gb: "$plan_info.data_gb"
    }
  }
]);
```

```
{
  subscriber_id: 1001,
  msisdn: '959420001001',
  name: 'Aung Aung',
  city: 'Yangon',
  plan_name: 'Data Starter',
  monthly_fee: 5000,
  data_gb: 5
}
{
  subscriber_id: 1002,
  msisdn: '959420001002',
  name: 'Su Su',
  city: 'Mandalay',
  plan_name: 'Business Plus',
  monthly_fee: 30000,
  data_gb: 50
}
{
  subscriber_id: 1003,
  msisdn: '959420001003',
  name: 'Mg Mg',
  city: 'Naypyidaw',
  plan_name: 'Youth Pack',
  monthly_fee: 10000,
  data_gb: 15
}
```

Exercise 5: Call Usage Summary by Caller

>_MONGOSH

```
> db.call_records.aggregate([
    {
      $group: {
        _id: "$caller_msisdn",
        total_duration_sec: { $sum: "$duration_sec" },
        total_cost: { $sum: "$cost" },
        total_calls: { $sum: 1 }
      }
    },
    { $sort: { total_cost: -1 } }
  ]);
< {
    _id: '959420001001',
    total_duration_sec: 180,
    total_cost: 570,
    total_calls: 2
  }
  {
    _id: '959420001002',
    total_duration_sec: 600,
    total_cost: 350,
    total_calls: 1
  }
  {
    _id: '959420001004',
    total_duration_sec: 90,
    total_cost: 80,
    total_calls: 1
  }
```

Exercise 6: Network Issue Report by City

```
> db.network_events.aggregate([
    {
      $group: {
        _id: "$city",
        total_events: { $sum: 1 },
        total_affected_users: { $sum: "$affected_users" }
      }
    },
    { $sort: { total_affected_users: -1 } }
  ]);
< {
    _id: 'Mandalay',
    total_events: 1,
    total_affected_users: 3000
  }
  {
    _id: 'Yangon',
    total_events: 2,
    total_affected_users: 1900
  }
```

Exercise 7: Unpaid Postpaid Billing Report

```
> db.billing.aggregate([
    { $match: { payment_status: "Unpaid" } },
    {
      $lookup: {
        from: "subscribers",
        localField: "subscriber_id",
        foreignField: "subscriber_id",
        as: "subscriber_info"
      }
    },
    { $unwind: "$subscriber_info" },
    {
      $project: {
        _id: 0,
        bill_id: 1,
        subscriber_id: 1,
        name: "$subscriber_info.name",
        msisdn: "$subscriber_info.msisdn",
        city: "$subscriber_info.city",
        total_amount: 1,
        payment_status: 1
      }
    }
  ]);
```

```
< {
    bill_id: 'B001',
    subscriber_id: 1002,
    total_amount: 44100,
    payment_status: 'Unpaid',
    name: 'Su Su',
    msisdn: '959420001002',
    city: 'Mandalay'
  }
```

Exercise 8: Update Suspended Subscriber

```
> db.subscribers.updateOne(
    { subscriber_id: 1003 },
    { $set: { status: "Active" } }
  );
```

```
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
```

Exercise 9: Delete Failed Recharge

```
> db.recharges.deleteMany({
    status: "Failed"
  });
< {
    acknowledged: true,
    deletedCount: 1
  }
```

Exercise 10: Create Indexes

```
> db.subscribers.createIndex({ subscriber_id: 1 }, { unique: true });
  db.subscribers.createIndex({ msisdn: 1 }, { unique: true });
  db.recharges.createIndex({ subscriber_id: 1 });
  db.call_records.createIndex({ caller_msisdn: 1 });
  db.call_records.createIndex({ call_start: -1 });
  db.billing.createIndex({ payment_status: 1 });
< payment_status_1
```

Exercise 11: Explain Query

```
>_MONGOSH
> db.call_records.find({
    caller_msisdn: "959420001001"
  }).explain("executionStats");
< {
    explainVersion: '1',
    queryPlanner: {
      namespace: 'telecom_lab_db.call_records',
      parsedQuery: {
        caller_msisdn: {
          '$eq': '959420001001'
        }
      },
      indexFilterSet: false,
      queryHash: 'A502E5E5',
      planCacheShapeHash: 'A502E5E5',
      planCacheKey: '34E221D3',
      optimizationTimeMillis: 0,
      maxIndexedOrSolutionsReached: false,
      maxIndexedAndSolutionsReached: false,
```

Exercise 12: Difficult Aggregation: Customer 360 View

```
>_MONGOSH
> db.subscribers.aggregate([
  {
    $lookup: {
      from: "plans",
      localField: "plan_id",
      foreignField: "plan_id",
      as: "plan"
    }
  },
  { $unwind: "$plan" },
  {
    $lookup: {
      from: "recharges",
      localField: "subscriber_id",
      foreignField: "subscriber_id",
      as: "recharge_history"
    }
  },
  {
    $lookup: {
      from: "billing",
      localField: "subscriber_id",
      foreignField: "subscriber_id",
      as: "billing_history"
    }
  },
```

```
  {
    $project: {
      _id: 0,
      subscriber_id: 1,
      name: 1,
      msisdn: 1,
      city: 1,
      status: 1,
      plan_name: "$plan.plan_name",
      total_success_recharge: {
        $sum: {
          $map: {
            input: {
              $filter: {
                input: "$recharge_history",
                as: "r",
                cond: { $eq: ["$$r.status", "Success"] }
              }
            },
            as: "sr",
            in: "$$sr.amount"
          }
        }
      },
      total_bill_amount: {
        $sum: "$billing_history.total_amount"
      }
    }
  }
]);
```

```
{
  subscriber_id: 1001,
  msisdn: '959420001001',
  name: 'Aung Aung',
  city: 'Yangon',
  status: 'Active',
  plan_name: 'Data Starter',
  total_success_recharge: 8000,
  total_bill_amount: 8925
}
{
  subscriber_id: 1002,
  msisdn: '959420001002',
  name: 'Su Su',
  city: 'Mandalay',
  status: 'Active',
  plan_name: 'Business Plus',
  total_success_recharge: 0,
  total_bill_amount: 44100
}
```

```
{
  subscriber_id: 1003,
  msisdn: '959420001003',
  name: 'Mg Mg',
  city: 'Naypyidaw',
  status: 'Active',
  plan_name: 'Youth Pack',
  total_success_recharge: 0,
  total_bill_amount: 0
}
{
  subscriber_id: 1004,
  msisdn: '959420001004',
  name: 'Hla Hla',
  city: 'Yangon',
  status: 'Active',
  plan_name: 'Data Starter',
  total_success_recharge: 10000,
  total_bill_amount: 0
}
```